

Aim: To generate and observe amplitude modulation (AM) using generator and DSO.

Objective: To generate amplitude modulated (AM) signal using a signal generator & to observe the modulated wave forms on a digital storage oscilloscope (DSO). The experiment also aims to study the effect of modulating signal on the carrier signal & to understand basic waveform parameters such as frequency, amplitude & waveform shape.

Theory: Amplitude modulation (AM) is a modulation technique in which the amplitude of higher frequency carrier signal is varied in accordance with the instantaneous amplitude of a low-frequency modulating signal, while the carrier frequency remains constant.

Definition

Frequency: It is defined as the number of cycles of a waveform completed per second. It is measured in Hertz (Hz).

Amplitude: It is the maximum value of signal measured from its mean (zero) level.

Waveform: A waveform is the shape of a signal as represented on a graph of voltage versus time.

Circuit Diagram

DSO Input

Signal Generator

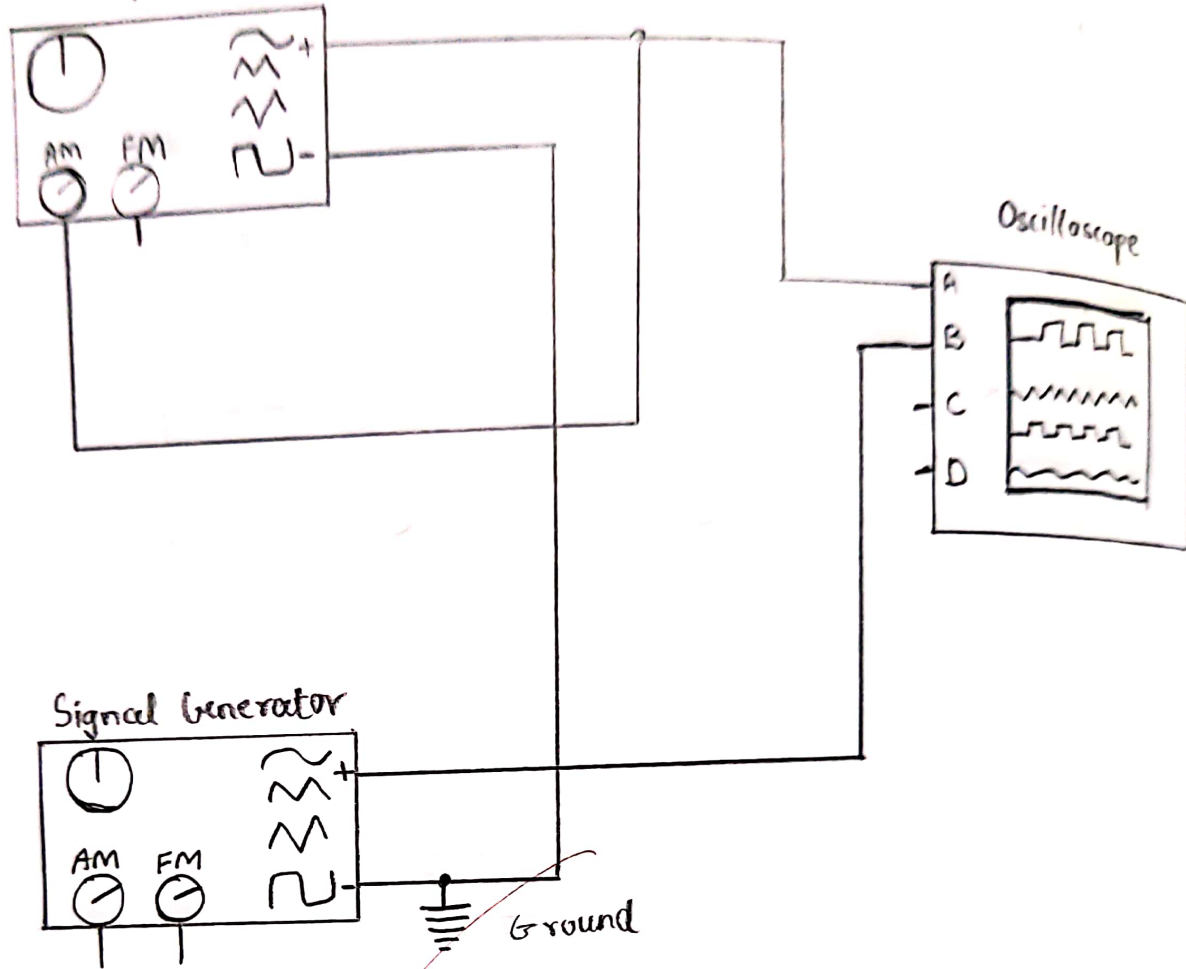


Fig 1.1

Working Principle:

1. One signal generator produces a high frequency carrier signal.
2. Another signal generator produces a low frequency modulating signal.
3. The carrier & modulating signals are combined to form an AM signal.
4. The AM output is fed to the DSO.
5. The DSO displays carrier, amplitude modulated waveform & modulating signal. The envelope of the AM signal clearly represents the modulating signal.

Observation:

- The carrier signal has the constant frequency but varying amplitude.
- The envelope of the AM waveform follows the shape of the modulating signal.
- The DSO accurately displays amplitude & frequency variation.
- Proper triggering ensures stable waveform display.

Conclusion:

Amplitude Modulation was successfully generated using a signal generator and observed on a digital storage Oscilloscope. The experiment confirms that in AM, the amplitude of the carrier signal varies in accordance with the modulating signal, while its frequency remains constant. The DSO output clearly shows the AM envelope verifying the theoretical concept of amplitude modulation.

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DSO output

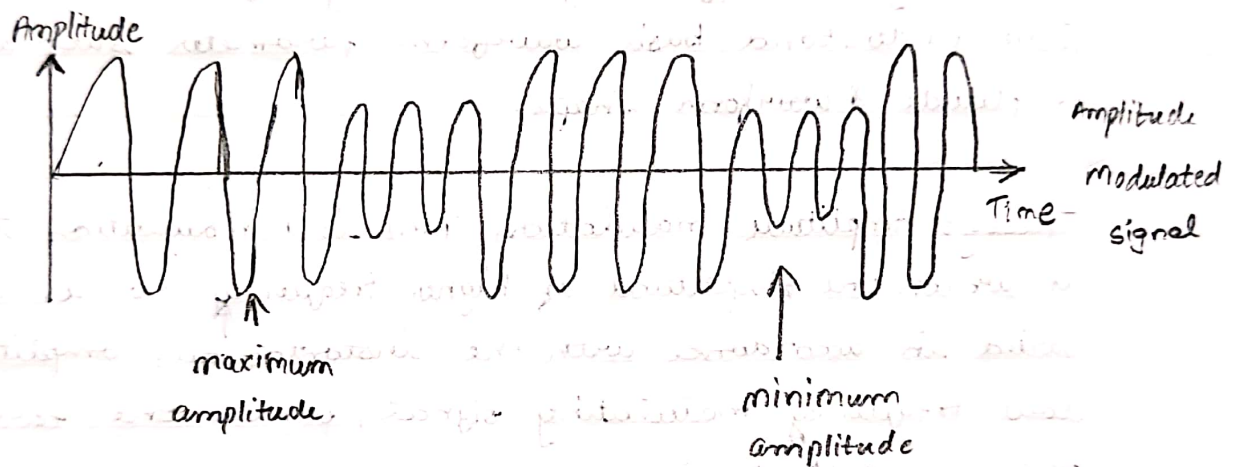
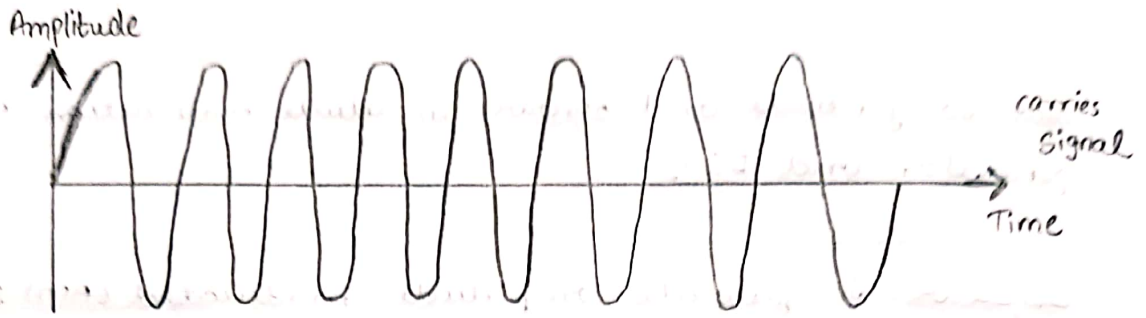
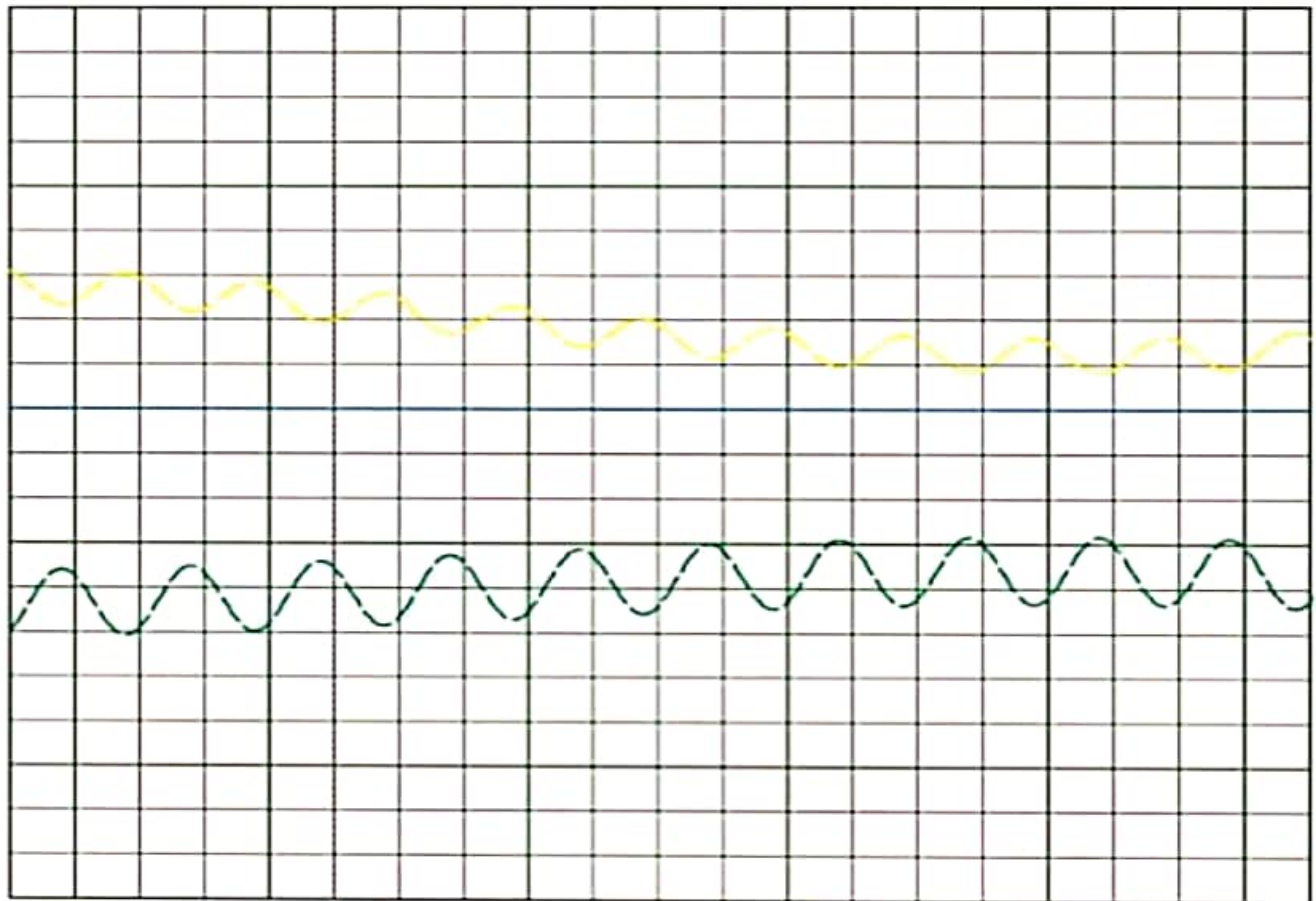


fig 1.2



				
	Channel A	Channel B	Channel C	Channel D
V/Div	5.00 V	5.00 V	5.00 V	5.00 V
Offset	60.00 V	20.00 V	-20.00 V	-60.00 V
Invert	Normal	Normal	Normal	Normal
Coupling	AC	AC	Off	AC
Source	Horizontal		Trigger	
Position	Trace		Channel A	
S/Div	5.00 mS		Level	
	1.00 mS		DC	
			Edge	
			Rising	
			Mode	
			Auto	